$$\min J\left[\overset{\mathrm{r}}{u}{}_{\xi\eta}^{p}(t),\overset{\mathrm{r}}{u}{}_{\xi\eta}^{n}(t),\overset{\mathrm{r}}{i}{}_{\xi\eta}^{p}(t),\overset{\mathrm{r}}{i}{}_{\xi\eta}^{n}(t),u_{g}^{dc}\right]$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0 & 1\\ -\omega_g & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}l_{u1}\\ l_{u2}\end{bmatrix}\left(x_1-\breve{x}_1\right)$$